

Linear Models: Probability and Linear Algebra Review

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Two Ways an Empirical Project Can Fail

1) Identification Failure

- The causal parameter (e.g. τ) is not uniquely determined by the observable data.
- Multiple causal stories are observationally equivalent.

Implication:

Your research question cannot be answered with these data.

2) Estimation / Inference Failure

- The causal parameter *is* determined by the data under your assumptions.
- But the estimator targets the wrong object or uncertainty is mismeasured.

Implication:

Your numerical answer or reported certainty may be wrong.

Example: Identification Is Fine, Estimation Is Not

- Suppose we run a block-randomized experiment,
- Treatment is assigned at the state level,
- Outcomes are observed at the county level.

Identification:

- Difference in means across counties identifies the ATE.

Estimation Problem:

- Outcomes are correlated within states.
- Naive OLS treats counties as independent.

Consequence:

- The coefficient is consistent.
- Standard errors can be severely understated (10x or more!).

The reported certainty may be false.

Goals for Today

- This is a condensed review of probability theory and linear algebra.
- We focus on the concepts that connect directly to econometric practice:
 - 1 Expectation, variance, and why they matter for prediction.
 - 2 Joint, marginal, and conditional distributions.
 - 3 The Conditional Expectation Function (CEF) as the target of regression.
 - 4 Matrix algebra and the geometry of projection.
 - 5 Positive definiteness, eigenvalues, and degrees of freedom.

Summarizing a Population

Suppose Y is a random variable with a known distribution. You need a single number μ to stand in for Y . Which number is “best”?

The answer depends on how you measure error, your **loss function**:

- **Squared-error loss** $(Y - \mu)^2 \Rightarrow$ best summary is $\mathbb{E}[Y]$ (the mean)
- **Absolute loss** $|Y - \mu| \Rightarrow$ best summary is the median

Why squared-error loss in this course? The minimized squared error *is* the variance:
 $\mathbb{E}[(Y - \mathbb{E}[Y])^2] = \text{Var}(Y)$.

Variance leads to covariance, covariance defines an inner product, and inner products give us orthogonal projection, these are the geometric foundation of OLS.

Next: we define $\mathbb{E}[Y]$, then prove it minimizes squared-error loss.

Expectation

Definition

Expected Value: define the expected value of Y as,

$$\mu = \mathbb{E}[Y] = \sum_{j=1} \tau_j P[Y = \tau_j] \quad \text{when } Y \text{ takes on discrete values } \tau$$

$$= \mathbb{E}[Y] = \int_{-\infty}^{\infty} y f(y) dy \quad \text{when } Y \text{ is continuous}$$

For all values of x with $p(x)$ greater than zero, take the sum/integral of values times the probability weights.

Unified notation (Riemann-Stieltjes): $\mathbb{E}[X] = \int_{-\infty}^{\infty} x dF(x)$

Key Properties of Expectation

The fact that expected values are sums/integrals gives us the following properties, for random variable X and Y , constant a .

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[a] = a$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

$$\mathbb{E}[\mathbb{E}[X]] = \mathbb{E}[X]$$

$$\mathbb{E}[XY] \neq \mathbb{E}[X] \times \mathbb{E}[Y] \quad (\text{in general})$$

Expectation Minimizes Mean Squared Prediction Error

Now we prove the claim: under squared-error loss, $\mathbb{E}[Y]$ is the best single-number summary of Y . Our prediction is μ ; minimize:

$$\begin{aligned} M &= \mathbb{E}[(Y - \mu)^2] \\ &= \mathbb{E}[Y^2] - 2\mu\mathbb{E}[Y] + \mu^2 \end{aligned}$$

Using calculus to minimize:

$$\begin{aligned} \frac{d}{d\mu} M &= -2\mathbb{E}[Y] + 2\mu = 0 \\ \mu^* &= \mathbb{E}[Y] \end{aligned}$$

This is a special case of the fact that the *conditional expectation function* minimizes mean-square prediction error.

Variance

Definition

The variance of a random variable X , $\text{var}(X)$, is

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2\end{aligned}$$

- Standard deviation: $\text{sd}(X) = \sqrt{\text{Var}(X)}$, population variance: σ^2 .

Corollary

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Sample Variance: Two Equivalent Forms

$$\begin{aligned}
 \widehat{\text{Var}}(X) &= \frac{1}{N} \sum_i (x_i - \bar{x})^2 \\
 &= \frac{1}{N} \sum_i (x_i - \bar{x})(x_i - \bar{x}) \\
 &= \frac{1}{N} \sum_i [(x_i - \bar{x})x_i - (x_i - \bar{x})\bar{x}] \\
 &= \frac{1}{N} \sum_i (x_i - \bar{x})x_i - \underbrace{\bar{x} \cdot \frac{1}{N} \sum_i (x_i - \bar{x})}_{=0} \\
 &= \frac{1}{N} \sum_i (x_i - \bar{x})x_i
 \end{aligned}$$

The key step: $\frac{1}{N} \sum_i (x_i - \bar{x}) = \bar{x} - \bar{x} = 0$.

This identity—that deviations from the mean sum to zero—will reappear when we derive OLS.

Covariance

Definition

The **covariance** of two random variables X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Key properties:

- $\text{Cov}(X, X) = \text{Var}(X)$
- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$
- If $X \perp Y$, then $\text{Cov}(X, Y) = 0$. The converse is false in general.

The Variance-Covariance Matrix

- Scalar case: $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$
- To generalize, we need a way to collect all variances and covariances in one object.
- For a random vector $\mathbf{X} = (X_1, \dots, X_k)'$, define the **variance-covariance matrix**:

$$\text{Var}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])'] = \Sigma$$

- Σ is $k \times k$, symmetric, and positive semi-definite.
- Diagonal entries are variances; off-diagonal entries are covariances.

Variance of Linear Combinations (Matrix Form)

- For any fixed matrix $\mathbf{A}$ ($m \times k$) and vector $\mathbf{b}$ ($m \times 1$):

$$\text{Var}(\mathbf{AX} + \mathbf{b}) = \mathbf{A} \text{Var}(\mathbf{X}) \mathbf{A}' = \mathbf{A} \Sigma \mathbf{A}'$$

- This formula appears throughout the course:
 - Variance of $\hat{\beta}$: $\text{Var}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}|\mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\sigma^2\mathbf{I}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$
 - Sandwich formula, GLS, robust standard errors—all follow this pattern.

Moments and Regularity Conditions

- $\mathbb{E}[X^r] = \int_{-\infty}^{\infty} x^r dF(x)$ is the r th *moment* of X .
- For some distributions, the expectation, the variance, or "higher" moments may not be finite.
- When $\mathbb{E}[X^r] = \infty$, the r th moment does not exist.
- Examples:
 - Fat tails distributions (e.g. Pareto distributions) often have no finite variance
 - Ratios: if X, Y are independent standard normal, $Z = X/Y$ has no finite expectation.
- Many econometric results require finite second (or fourth) moments to offer probabilistic guarantees.
- Feel free for this class to assume all moments exist.

The Normal Distribution

Definition

$X \sim \text{Normal}(\mu, \sigma^2)$ has the probability density function

$$f(x) = \frac{1}{\sqrt{2\sigma^2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Key properties:

- If $X \sim N(\mu, \sigma^2)$, then $aX + b \sim N(a\mu + b, a^2\sigma^2)$.
- If $X_1 \perp X_2$ and both normal, $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.
- If X_1 and X_2 are *jointly* normal and uncorrelated, then they are independent.

Moments of the Normal Distribution

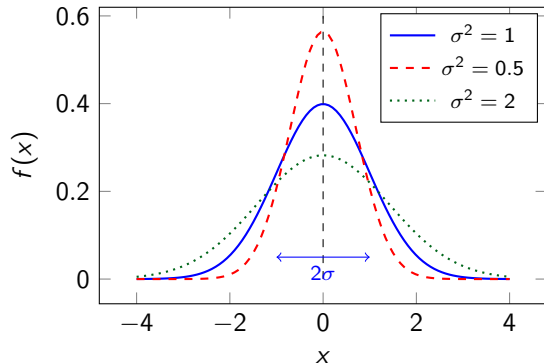
Let $X \sim N(\mu, \sigma^2)$.

Raw Moments

- First moment: $\mathbb{E}[X] = \mu$
- Second moment: $\mathbb{E}[X^2] = \sigma^2 + \mu^2$

Central Moments

- Variance: $\mathbb{E}[(X - \mu)^2] = \sigma^2$
- Third: $\mathbb{E}[(X - \mu)^3] = 0$ (symmetric)
- Fourth: $\mathbb{E}[(X - \mu)^4] = 3\sigma^4$



What Is a Parameter?

A **parameter** is a fixed, unknown number that summarizes something about the population — for example, a mean, a variance, or a regression slope.

We have already seen one way to define a parameter: as the solution to an optimization problem (e.g. $\mu^* = \arg \min \mathbb{E}[(Y - \mu)^2]$).

There is an equivalent approach: define a parameter as the value that makes a certain population expectation equal zero. These are called **moment conditions**.

Moment Conditions: Examples

Example 1: Mean

$$\mathbb{E}[X - \mu] = 0.$$

The true μ is the unique value that makes this expectation zero.

Example 2: Linear Regression slope

$$\mathbb{E}[X(Y - X'\beta)] = 0.$$

The true β is the value that makes the regressors X uncorrelated with the residual $(Y - X'\beta)$.

In both cases, the parameter is *defined* by the condition — not estimated from data (yet).

General Form of Method of Moments

Many models can be written as:

$$\mathbb{E}[g(W, \theta)] = 0.$$

- W = observable data.
- θ = parameter.
- $g(\cdot)$ encodes the economic or statistical restrictions.

Identification requires that these conditions determine a unique θ .

Roadmap: From Joint Distributions to the CEF

So far we have summarized a single random variable Y using $\mathbb{E}[Y]$.

But in practice we want to summarize Y *given information about X* . We want $\mathbb{E}[Y|X = x]$, the **conditional expectation function** (CEF).

To get there, we need three building blocks:

- 1 **Joint distributions**: how X and Y vary together.
- 2 **Marginal distributions**: the distribution of X alone.
- 3 **Conditional distributions**: the distribution of Y for a fixed value of X .

Once we have the conditional distribution, the CEF is just its expectation, and it will be the *target* that regression approximates.

Joint Distributions

Recall,

Definition

The joint distribution function of (X, Y) is $F(x, y) = P[X \leq x, Y \leq y]$.

Definition

The joint density is $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$.

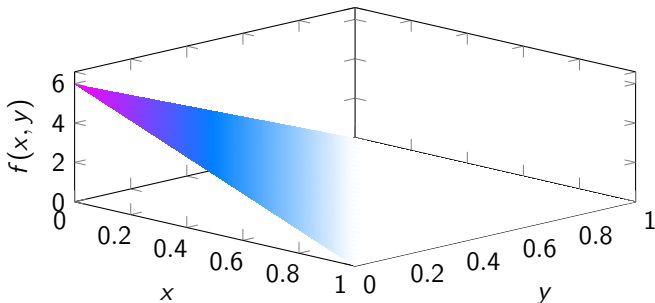
Joint densities and distributions live in 3 (or higher) dimensions.

Example: A Joint Density

Let X and Y be the fractions of the electorate mobilized by parties A and B, so $X, Y \geq 0$ and $X + Y \leq 1$.

$$f(x, y) = 6(1 - x - y)$$

The density is highest near $(0, 0)$: low mobilization by both parties is the most common outcome (6 is a normalizing constant).

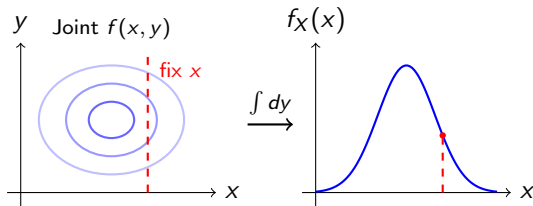


Marginal Density

Definition

The **marginal density** of X is $f_X(x) = \underbrace{\int_{-\infty}^{\infty} f(x, y) dy}_{\text{integrate out } y}$

Fix x and sum the joint density over all possible y . The result is a function of x alone.



Conditional Density

Definition

The **conditional density** of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{\overbrace{f(x, y)}^{\text{joint}}}{\underbrace{f_X(x)}_{\text{marginal}}}$$

for any x such that $f_X(x) > 0$.

Take a slice of the joint density at a fixed x , then rescale so it integrates to 1. This is the distribution of Y for units with that particular value of X .

Example: Marginal Density from $f(x, y) = 6(1 - x - y)$

Marginal density of X : fix x , integrate out y .

What are the limits of integration? The density is positive when $x, y \geq 0$ and $x + y \leq 1$, so for a given x :

- Lower bound: $y = 0$ (mobilization cannot be negative).
- Upper bound: $y = 1 - x$ (the constraint $x + y \leq 1$).

$$f_X(x) = \int_{\underbrace{0}_{\min y}}^{\underbrace{1-x}_{\max y \text{ given } x}} 6(1 - x - y) dy = 6 \left[(1-x)y - \frac{y^2}{2} \right]_0^{1-x} = 3(1-x)^2$$

Example: Conditional Density from $f(x, y) = 6(1 - x - y)$

Conditional density of $Y|X = x$: divide the joint by the marginal.

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{6(1 - x - y)}{3(1 - x)^2} = \frac{2(1 - x - y)}{(1 - x)^2}$$

For each fixed x , this is a valid density in y on $[0, 1 - x]$.

Note the effect of conditioning: the *shape* of the distribution of Y changes depending on the value of X . When x is large, the remaining “room” for y is small.

Conditional Expectation

$$\begin{aligned}\mathbb{E}[Y|X = x] &= \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy \\ &= \frac{\int_{-\infty}^{\infty} y f(y, x) dy}{\int_{-\infty}^{\infty} f(y, x) dy}\end{aligned}$$

The average value of Y given that X equals the specific value x .

Conditional Expectation Function (CEF)

■ CEF:

$$\mathbb{E}[Y|X = x] = m(x)$$

- Y is the dependent variable, $X = (X_1, \dots, X_k)'$ are the independent variables.
- $m(x) = \mathbb{E}[Y|X = x]$ is the value of a function at the real valued vector x .
- $m(X) = \mathbb{E}[Y|X]$ is a function of random variables, so is itself a random variable.
- We will show that the CEF is the best predictor of Y given X in the mean-square error sense.
- In most applications we use a *linear approximation* to the CEF, then make inferences about the joint distribution.

Example: Computing the CEF from $f(x, y) = 6(1 - x - y)$

Apply the definition using our conditional density:

$$\begin{aligned}\mathbb{E}[Y|X = x] &= \int_0^{1-x} y f_{Y|X}(y|x) dy \\&= \int_0^{1-x} y \cdot \frac{2(1-x-y)}{(1-x)^2} dy \\&= \frac{2}{(1-x)^2} \int_0^{1-x} [(1-x)y - y^2] dy \\&= \frac{2}{(1-x)^2} \left[\frac{(1-x)y^2}{2} - \frac{y^3}{3} \right]_0^{1-x} \\&= \frac{2}{(1-x)^2} \left[\frac{(1-x)^3}{2} - \frac{(1-x)^3}{3} \right] = \frac{2}{(1-x)^2} \cdot \frac{(1-x)^3}{6} \\&= \frac{1-x}{3}\end{aligned}$$

Law of Iterated Expectations

Theorem

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y],$$

$$\begin{aligned}\mathbb{E}[\mathbb{E}[Y|X]] &= \int_{-\infty}^{\infty} \mathbb{E}[Y|X=x] f_X(x) dx \\&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy f_X(x) dx \\&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \frac{f(y,x)}{f_X(x)} f_X(x) dy dx \\&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(y,x) dy dx \\&= \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}[Y]\end{aligned}$$

Law of Total Variance

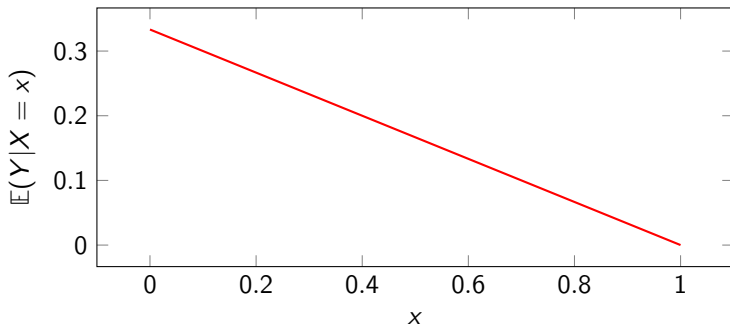
Theorem

$$\text{Var}[Y] = \mathbb{E}[\text{Var}[Y|X]] + \text{Var}[\mathbb{E}[Y|X]]$$

- $\text{Var}[\mathbb{E}[Y|X]]$: variance of the CEF — the “explained” variance.
- $\mathbb{E}[\text{Var}[Y|X]]$: average residual variance — the “unexplained” variance.
- This decomposition underpins R^2 : the fraction of the total variance of Y explained by X .
- You should be able to prove and apply both the LIE and LTV.

Example: CEF from a Joint Density

Joint density: $f(x, y) = 6(1 - x - y)$ for $x \geq 0$, $y \geq 0$, $x + y \leq 1$. Marginal: $f_X(x) = 3(1 - x)^2$. CEF: $\mathbb{E}(Y|X = x) = \frac{1-x}{3}$.



As X increases, $\mathbb{E}[Y|X]$ decreases linearly—this CEF *is* linear.

Roadmap: Why Matrices?

We now have the CEF, $\mathbb{E}[Y|X]$, as our target. In most applications we approximate it with a *linear* function of X .

With one regressor this is just a line. But with k regressors we need a compact way to write and manipulate systems of linear equations. That is what matrix algebra provides.

This section reviews the tools we will use repeatedly:

- 1 Matrix notation:** writing $\mathbf{y} = \mathbf{X}\beta$ instead of n separate equations.
- 2 Key operations:** transpose, inverse, rank.
- 3 Projection:** the geometric interpretation of OLS as an orthogonal projection.

Linear Systems and Matrix Representation

A system of linear equations

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots$$

$$y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots$$

$$\vdots$$

can be written compactly as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$$

where $\mathbf{y}$ is $n \times 1$, $\mathbf{X}$ is $n \times k$, and $\boldsymbol{\beta}$ is $k \times 1$.

Matrix Multiplication

If $\mathbf{A}$ is $k \times r$ and $\mathbf{B}$ is $r \times s$, they are **conformable** and

$$(\mathbf{AB})_{ij} = \sum_{\ell=1}^r a_{i\ell} b_{\ell j}$$

The result is $k \times s$.

Key rules:

- Generally $\mathbf{AB} \neq \mathbf{BA}$.
- $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ (distributive).
- $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ (associative).
- $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$, where $'$ is the transpose.

Inner Product and Similarity

- The inner product of two $k \times 1$ vectors:

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}'\mathbf{b} = \sum_{j=1}^k a_j b_j$$

- Compare to the sample covariance for demeaned variables:

$$\text{Cov}(\mathbf{x}, \mathbf{y}) = \frac{1}{n-1} \sum_{i=1}^n x_i y_i$$

- Two vectors are **orthogonal** if $\mathbf{a}'\mathbf{b} = 0$.
- $\|\mathbf{a}\| = \sqrt{\mathbf{a}'\mathbf{a}}$ is the Euclidean norm (length) of $\mathbf{a}$.

The Design Matrix $\mathbf{X}$

In regression, $\mathbf{X}$ is the $n \times k$ **design matrix**: rows are observations, columns are variables.

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,k-1} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,k-1} \end{bmatrix} = \begin{bmatrix} \mathbf{x}'_1 \\ \mathbf{x}'_2 \\ \vdots \\ \mathbf{x}'_n \end{bmatrix} = [\mathbf{i} \quad \mathbf{c}_1 \quad \mathbf{c}_2 \quad \cdots \quad \mathbf{c}_{k-1}]$$

- The first column $\mathbf{i} = (1, 1, \dots, 1)'$ is the intercept.
- Each row $\mathbf{x}'_i$ is observation i 's vector of regressors.
- Each column $\mathbf{c}_j$ is the $n \times 1$ vector of all observations on variable j .
- $\mathbf{X}$ has n rows (observations) and k columns (independent variables+intercept).

Matrix Inverse

- If a $k \times k$ matrix $\mathbf{A}$ is *nonsingular* (full rank), there exists a unique $\mathbf{A}^{-1}$ such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_k$.
- Key formulas:
 - $(\mathbf{A}^{-1})' = (\mathbf{A}')^{-1}$
 - $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- The *rank* of a matrix is the number of linearly independent columns.
- A matrix is singular (non-invertible) when its columns are linearly dependent—this corresponds to **perfect multicollinearity** in regression.

Quadratic Forms: An Example

- We often want to characterize a 2nd degree polynomial like

$$3x_1^2 + 4x_2^2 + 9x_3^2 - 5x_1x_3.$$

- We can write it as a quadratic form:

$$\mathbf{x}^\top \mathbf{A} \mathbf{x}.$$

- The diagonal elements of $\mathbf{A}$ are the coefficients on x_i^2 .
- Cross terms are split across symmetric entries:

$$-5x_1x_3 \Rightarrow a_{13} = a_{31} = -\frac{5}{2}.$$

- Thus,

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & -\frac{5}{2} \\ 0 & 4 & 0 \\ -\frac{5}{2} & 0 & 9 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Positive Definiteness

A **quadratic form** is a scalar $\mathbf{x}'\mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is symmetric: $\mathbf{x}'\mathbf{A}\mathbf{x} = \sum_i a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j$. The sign of this scalar classifies the matrix:

Definition

A symmetric matrix $\mathbf{A}$ is **positive definite** (PD) if $\mathbf{c}'\mathbf{A}\mathbf{c} > 0$ for every $\mathbf{c} \neq \mathbf{0}$. It is **positive semi-definite** (PSD) if $\mathbf{c}'\mathbf{A}\mathbf{c} \geq 0$ for every $\mathbf{c} \neq \mathbf{0}$.

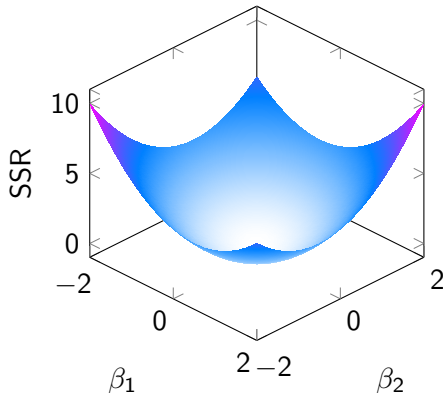
Intuition: PD means the quadratic form is always positive—no matter which direction $\mathbf{c}$ you choose, the “bowl” curves upward.

Why Positive Definiteness Matters

- Variance-covariance matrices are positive semi-definite by construction.
- If $\mathbf{X}$ has full column rank, $\mathbf{X}'\mathbf{X}$ is positive definite, guaranteeing a unique OLS solution.
- PD allows us to compare matrices (which estimator has “smaller” variance):

$\mathbf{A} - \mathbf{B}$ is PD $\implies \mathbf{A}$ is “larger” than $\mathbf{B}$

- A PD quadratic form is strictly convex with a unique global minimum—the OLS “bowl.”



How to Check Positive Definiteness

Practical tests:

- 1 **Eigenvalues:** $\mathbf{A}$ is PD $\iff$ all eigenvalues are positive.
- 2 **Cholesky decomposition:** $\mathbf{A}$ is PD $\iff$ it can be factored as $\mathbf{A} = \mathbf{L}\mathbf{L}'$ for a lower-triangular $\mathbf{L}$ with positive diagonal. This is how software checks PD in practice.
- 3 **In regression:** if $\mathbf{X}$ is $n \times k$ with full column rank, then $\mathbf{X}'\mathbf{X}$ is PD. So the practical question is: are your regressors linearly independent?

Properties (reference):

- If $\mathbf{A}$ is PD, it is nonsingular and $\mathbf{A}^{-1}$ is also PD.
- If $\mathbf{A}$ and $\mathbf{B}$ are PD, so is $\mathbf{A} + \mathbf{B}$.
- If $\mathbf{A}$ is PD and $c > 0$, then $c\mathbf{A}$ is PD.

The Gram Matrix $\mathbf{X}'\mathbf{X}$

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} \sum x_{1i}^2 & \sum x_{1i}x_{2i} & \cdots \\ \sum x_{1i}x_{2i} & \sum x_{2i}^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- $\mathbf{X}'\mathbf{X}$ is symmetric and positive semi-definite.
- If we normalize columns so $\|\mathbf{x}_j\| = 1$, then $(\mathbf{X}'\mathbf{X})_{ij} = \cos \theta_{ij}$: a measure of similarity.
- $\mathbf{X}'\mathbf{X}$ encodes all the second-moment information about the regressors.

Linear Combinations

Given vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$,
a **linear combination** is any vector of the form

$$c_1 \mathbf{x}_1 + \dots + c_k \mathbf{x}_k.$$

Example in $\mathbb{R}^2$:

- One vector: all multiples lie on a line.
- Two non-collinear vectors: combinations fill the plane.

Linear combinations describe what vectors you can “build.”

Span

The **span** of $\mathbf{x}_1, \dots, \mathbf{x}_k$ is

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{c_1\mathbf{x}_1 + \dots + c_k\mathbf{x}_k\}.$$

It is the set of *all* vectors you can build from them.

Geometric intuition:

- One independent vector $\Rightarrow$ a line.
- Two independent vectors $\Rightarrow$ a plane.
- Three independent vectors in $\mathbb{R}^3 \Rightarrow$ all of $\mathbb{R}^3$.

Linear Independence and Basis

Vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are **linearly independent** if

$$c_1 \mathbf{x}_1 + \dots + c_k \mathbf{x}_k = \mathbf{0}$$

implies

$$c_1 = \dots = c_k = 0.$$

Intuition:

- No vector can be written as a combination of the others.
- No redundancy.

If independent vectors span a space $\mathcal{V}$, they form a **basis**.

Basis Examples in $\mathbb{R}^2$

Standard basis: $\{(1, 0), (0, 1)\}$ is a basis for $\mathbb{R}^2$.

Another basis: $\{(1, 1), (-1, 1)\}$ is also a basis for $\mathbb{R}^2$.

Proof: Any $(x, y) \in \mathbb{R}^2$ can be written as $\alpha(1, 1) + \beta(-1, 1)$:

$$x = \alpha - \beta$$

$$y = \alpha + \beta$$

Adding and subtracting: $\alpha = \frac{x+y}{2}$, $\beta = \frac{y-x}{2}$.

So every vector has a representation (*spanning*). If $(x, y) = (0, 0)$, then $\alpha = \beta = 0$, so the set is *linearly independent*. Hence it is a basis.

Non-Basis Examples

Example 1: $\{(1, 0, 0), (0, 1, 0)\}$ is **not** a basis of $\mathbb{R}^3$.

- These vectors are linearly independent, but they do not span $\mathbb{R}^3$ —e.g. $(0, 0, 1)$ cannot be written as a linear combination.

Example 2: $\{(1, 0), (0, 1), (1, 1)\}$ is **not** a basis of $\mathbb{R}^2$.

- These vectors span $\mathbb{R}^2$, but they are linearly dependent:

$$1 \cdot (1, 0) + 1 \cdot (0, 1) + (-1) \cdot (1, 1) = (0, 0).$$

- A basis requires *both* spanning and linear independence.

Takeaway: A basis is the minimal spanning set (or equivalently, a maximal independent set).

Dimension and the Column Space

Dimension

The dimension of a space is the number of vectors in any basis.

In Regression

- The columns of $\mathbf{X}$ are your regressors.
- Their span is the **column space**.
- OLS projects $\mathbf{y}$ onto this space.
- Linear dependence $\Rightarrow$ no unique solution.

Full column rank = regressors are linearly independent.

Projection: The Geometric Heart of OLS

- **Projection Theorem:** Let $\mathcal{W}$ be a subspace. There exists a unique $\hat{\mathbf{y}} \in \mathcal{W}$ closest to $\mathbf{y}$:

$$\hat{\mathbf{y}} = \text{proj}_{\mathcal{W}}(\mathbf{y})$$

- **Projection onto a line:** If $\mathcal{W} = \{c \mathbf{x} : c \in \mathbb{R}\}$, then

$$\hat{\mathbf{y}} = \frac{\mathbf{x}'\mathbf{y}}{\mathbf{x}'\mathbf{x}} \mathbf{x}$$

- **General case:** If $\mathcal{W}$ is the column space of $\mathbf{X}$, then

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \equiv \mathbf{P}\mathbf{y}$$

where $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the **hat matrix**.

Orthogonality of Residuals

The error $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to every column of $\mathbf{X}$:

$$\begin{aligned}\mathbf{X}'(\mathbf{y} - \hat{\mathbf{y}}) &= \mathbf{X}'(\mathbf{y} - \mathbf{P}\mathbf{y}) \\ &= \mathbf{X}'\mathbf{y} - \underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'}_I \mathbf{X}'\mathbf{y} \\ &= \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{y} \\ &= \mathbf{0}\end{aligned}$$

- This is the matrix form of the OLS first-order conditions.
- Geometrically: the residual vector is perpendicular to the column space of $\mathbf{X}$.
- This is why $\sum_i \hat{e}_i = 0$ when there is an intercept (residuals are orthogonal to the column of ones).

Summarizing Matrices

There are several useful summaries of a matrix $\mathbf{A}$:

- **Trace:**

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n A_{ii}$$

- **Determinant:** $\det(\mathbf{A})$ measures (signed) volume scaling; $\det(\mathbf{A}) = 0$ iff $\mathbf{A}$ is singular.
- **Rank:** $\text{rank}(\mathbf{A})$ is the dimension of the column space.
- **Eigenvalues:** $\lambda_1, \dots, \lambda_n$ summarize stretching along special directions.

We will use these repeatedly to diagnose invertibility and curvature.

Properties of the Trace

The trace has several properties that we will use repeatedly:

- 1 $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$ (linearity)
- 2 $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$ (linearity)
- 3 $\text{tr}(\mathbf{A}^T) = \text{tr}(\mathbf{A})$ (transpose invariance)
- 4 $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ (cyclic property)

Proof of (4): $\text{tr}(\mathbf{AB}) = \sum_i (\mathbf{AB})_{ii} = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \sum_j (\mathbf{BA})_{jj} = \text{tr}(\mathbf{BA})$

Where this matters: When we prove that $s^2 = \frac{\mathbf{e}'\mathbf{e}}{n-k}$ is unbiased for σ^2 , the key step uses

$$\mathbb{E}[\mathbf{e}'\mathbf{e}|\mathbf{X}] = \mathbb{E}\left[\underbrace{\text{tr}(\mathbf{M}\mathbf{e}\mathbf{e}')}_{\text{scalar=its own trace}} \middle| \mathbf{X} \right] = \text{tr}(\mathbf{M} \mathbb{E}[\mathbf{e}\mathbf{e}'|\mathbf{X}]) = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n-k)$$

Eigenvalues and Eigenvectors

- For a square matrix $\mathbf{A}$, if

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$$

for some nonzero $\mathbf{u}$, then $\mathbf{u}$ is an **eigenvector** and λ is the corresponding **eigenvalue**.

- Interpretation: along direction $\mathbf{u}$, the matrix acts like multiplication by λ .
- If $\mathbf{A}$ is symmetric, its eigenvalues are real and it has an orthonormal eigenbasis.
- Two useful identities:

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i \quad \text{and} \quad \det(\mathbf{A}) = \prod_{i=1}^n \lambda_i.$$

- Eigenvalues diagnose key properties (symmetric $\mathbf{A}$):
 - $\lambda_i > 0$ for all $i \iff \mathbf{A}$ is positive definite.
 - Some $\lambda_i = 0 \iff \mathbf{A}$ is singular.

Idempotent Matrices

- A matrix is **idempotent** if

$$\mathbf{A}^2 = \mathbf{A}.$$

Theorem

If $\mathbf{A}$ is idempotent, then all eigenvalues of $\mathbf{A}$ are 0 or 1.

- **Proof (one line):** If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, then

$$\mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x} \quad \text{but also} \quad \mathbf{A}^2\mathbf{x} = \mathbf{A}\mathbf{x} = \lambda\mathbf{x},$$

so $\lambda^2 = \lambda$, hence $\lambda \in \{0, 1\}$.

- The hat matrix

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

is symmetric and idempotent.

- $\text{rank}(\mathbf{P})$ equals the number of eigenvalues equal to 1 (so $\text{rank}(\mathbf{P}) = k$).
- Therefore $\text{tr}(\mathbf{P}) = k$.

Orthogonal Complements

Definition

The **orthogonal complement** of a subspace $\mathcal{W} \subset \mathbb{R}^n$ is

$$\mathcal{W}^\perp = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}'\mathbf{w} = 0 \text{ for all } \mathbf{w} \in \mathcal{W}\}.$$

In words: $\mathcal{W}^\perp$ is the set of all vectors orthogonal to every vector in $\mathcal{W}$.

Examples:

- If $\mathcal{W}$ is a line through the origin in $\mathbb{R}^2$, then $\mathcal{W}^\perp$ is the perpendicular line.
- If $\mathcal{W}$ is a plane through the origin in $\mathbb{R}^3$, then $\mathcal{W}^\perp$ is the normal line.
- In regression: $\text{col}(\mathbf{X})^\perp$ is the space where the residuals live.

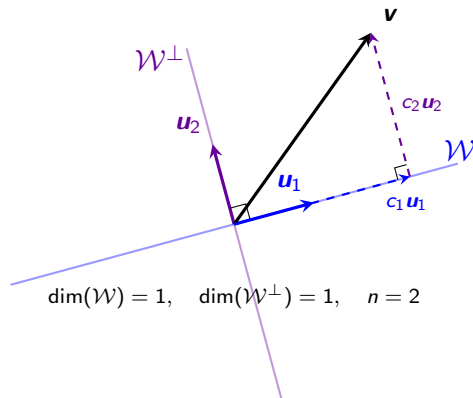
Fundamental Theorem: Dimension

Theorem (Fundamental Thm of Linear Algebra)

If $\mathcal{W}$ is a subspace of $\mathbb{R}^n$, then

$$\dim(\mathcal{W}) + \dim(\mathcal{W}^\perp) = n.$$

Proof setup: Let $\dim(\mathcal{W}) = k$. Choose an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_k$ for $\mathcal{W}$. Extend to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ for $\mathbb{R}^n$.



Proof: $\dim(\mathcal{W}) + \dim(\mathcal{W}^\perp) = n$

Claim: $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ form a basis for $\mathcal{W}^\perp$, so $\dim(\mathcal{W}^\perp) = n - k$.

Step 1 (orthogonality): For $j > k$ and $i \leq k$, orthonormality gives $\mathbf{u}_j' \mathbf{u}_i = 0$, hence $\mathbf{u}_j \in \mathcal{W}^\perp$.

Step 2 (spanning): Take any $\mathbf{v} \in \mathcal{W}^\perp$ and expand in the full basis:

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{u}_i.$$

For any $j \leq k$,

$$0 = \mathbf{v}' \mathbf{u}_j = c_j,$$

so $\mathbf{v} = \sum_{i=k+1}^n c_i \mathbf{u}_i$, which lies in $\text{span}\{\mathbf{u}_{k+1}, \dots, \mathbf{u}_n\}$.

Step 3 (independence): $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ are orthonormal, hence linearly independent.

Therefore $\dim(\mathcal{W}^\perp) = n - k$, so $\dim(\mathcal{W}) + \dim(\mathcal{W}^\perp) = k + (n - k) = n$. $\square$

Degrees of Freedom

Let $\mathcal{W} = \text{col}(\mathbf{X})$ be the column space of $\mathbf{X}$.

- $\dim(\mathcal{W}) = k$ (number of linearly independent regressors).
- The fitted values satisfy $\hat{\mathbf{y}} \in \mathcal{W}$.
- The residuals satisfy $\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}} \in \mathcal{W}^\perp$.
- By $\dim(\mathcal{W}) + \dim(\mathcal{W}^\perp) = n$,

$$\dim(\mathcal{W}^\perp) = n - k.$$

- This is why we divide by $n - k$ when estimating σ^2 : residual variation lives in an $(n - k)$ -dimensional space.

What Is the Derivative of a Vector Function?

Let $f(a_1, a_2, a_3, \dots, a_k) = f(\mathbf{a})$ be a scalar function of a vector $\mathbf{a} \in \mathbb{R}^k$.

The **gradient** is:

$$\nabla_{\mathbf{a}} f(\mathbf{a}) = \begin{bmatrix} \frac{\partial f}{\partial a_1} \\ \vdots \\ \frac{\partial f}{\partial a_k} \end{bmatrix}.$$

Key idea:

- Derivative of a scalar w.r.t. a vector is a vector.
- First-order condition for a minimum:

$$\nabla_{\mathbf{a}} f(\mathbf{a}) = \mathbf{0}.$$

Linear Forms

Let $f(\mathbf{a}) = \mathbf{z}'\mathbf{a}$, where $\mathbf{z}$ is fixed.

Write it out:

$$f(\mathbf{a}) = \sum_{j=1}^k z_j a_j.$$

Taking derivatives componentwise:

$$\nabla_{\mathbf{a}}(\mathbf{z}'\mathbf{a}) = \mathbf{z}.$$

More generally, if $f(\mathbf{a}) = \mathbf{Z}\mathbf{a}$,

$$\frac{d \mathbf{Z}\mathbf{a}}{d \mathbf{a}} = \mathbf{Z}.$$

Derivatives of Quadratic Forms

Let

$$f(\mathbf{a}) = \mathbf{a}'\mathbf{Z}\mathbf{a}.$$

Write it out:

$$f(\mathbf{a}) = \sum_{i,j} a_i Z_{ij} a_j.$$

Taking derivatives (each a_ℓ appears as both a_i and a_j , giving two terms):

$$\nabla_{\mathbf{a}}(\mathbf{a}'\mathbf{Z}\mathbf{a}) = (\mathbf{Z} + \mathbf{Z}')\mathbf{a}.$$

If $\mathbf{Z}$ is symmetric, so $\mathbf{Z} = \mathbf{Z}'$:

$$\nabla_{\mathbf{a}}(\mathbf{a}'\mathbf{Z}\mathbf{a}) = 2\mathbf{Z}\mathbf{a}.$$

Application: Deriving OLS

Least squares solves:

$$\min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta).$$

Expand (the cross terms $-\mathbf{y}'\mathbf{X}\beta$ and $-\beta'\mathbf{X}'\mathbf{y}$ are both 1×1 scalars, so they equal their own transpose and combine):

$$= \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta.$$

Take gradient using our linear and quadratic form rules, set equal to zero:

$$-2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\beta = \mathbf{0}.$$

Pre-multiply both sides by $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\Rightarrow \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Second-Order Condition: Verifying a Minimum

The FOC gave us a critical point. Is it a minimum?

The **Hessian** (matrix of second derivatives) of $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ is:

$$\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} [\mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}] = 2\mathbf{X}'\mathbf{X}$$

- If $\mathbf{X}$ has full column rank, then $\mathbf{X}'\mathbf{X}$ is positive definite (from our earlier result).
- A positive definite Hessian means the objective is strictly convex — the critical point is a unique global minimum.
- This connects two sections: positive definiteness guarantees both that $(\mathbf{X}'\mathbf{X})^{-1}$ exists *and* that the solution is a minimum.

Summary and Roadmap

- The **CEF** $\mathbb{E}[Y|\mathbf{X} = \mathbf{x}]$ is the target of regression—the best MSE predictor.
- **OLS** is the linear approximation: a projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$.
- The OLS residual is orthogonal to $\mathbf{X}$ (first-order conditions).
- Positive definiteness of $\mathbf{X}'\mathbf{X}$ guarantees existence and uniqueness.
- Degrees of freedom ($n - k$) come from the rank-nullity theorem.
- **Next:** the CEF vs Best Linear Predictor